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CSC 301 — Past Questions & Materials
by Renance

Course: CSC 301
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 1
Levels: 300

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Paper 1 of 1 — CSC 301 Data Structure

Level 300 • First Semester • PastQuestion

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**FEDERAL UNIVERSITY OF PETROLEUM RESOURCES,
EFFURUN**
DEPARTMENT OF COMPUTER SCIENCE
1ST SEMESTER BSc EXAMINATION (2025/2026 SESSION)
CSC301: DATA STRUCTURES

INSTRUCTIONS: Date: 24th March 2026

1. Attempt **Question 1** and any other 3.
2. Time allowed: **2½ hours**.
3. Answer each question on a **new page**.

Question 1

- a) With the aid of a diagram define data structure. (4mks)
- b) Enumerate three (3) reasons for organizing data. (3mks)
- c) Explain the following terms (i) Contiguous data structure (ii) Non-Contiguous data structure (iii) Primitive data structure (iv) Non-Primitive data structure. (8mks)
- d) List and explain four (4) advantages of Abstract Data types. (5mks)
- e) Given the list below in a Linked List, illustrate
 - (i) Insertion of a new node called "J" between the 3rd and 3rd element.
 - (ii) Deleting node "M" from the List.



(5mks)

Question 2

- a) With the aid of a diagram, explain the different types of Linked List. (5mks)
- b) Explain the term Stack and the two basic operations associated with it. (5mks)
- c) List the operations of an Array and use a line or two lines of syntax to implement one of the operation mentioned. (5mks)

Question 3

- a) Explain the term Queue. (2mks)
- b) Enumerate four (4) application of a queue. (4mks)
- c) State three (3) advantages and one (1) disadvantage of a Linked List. (4mks)
- d) Define Stack overflow condition and Stack underflow condition. (5mks)

Question 4

- a) What is an ArrayList? (3mks)
- b) Mention two (2) limitations of an ArrayList. (3mks)
- c) With the aid of a diagram, explain the term Linkrd List. (5mks)
- d) State five (5) applications of queue. (5mks)

Question 5

- a) List and explain two (2) classes on Non-Primitive data structure. (2mks)
- b) List three (3) steps for selecting data structure. (3mks)
- c) By means of suitable example, explain the term Array. (5mks)
- d) State three (3) advantages and two (2) disadvantages of an Array. (5mks)

Goodluck!